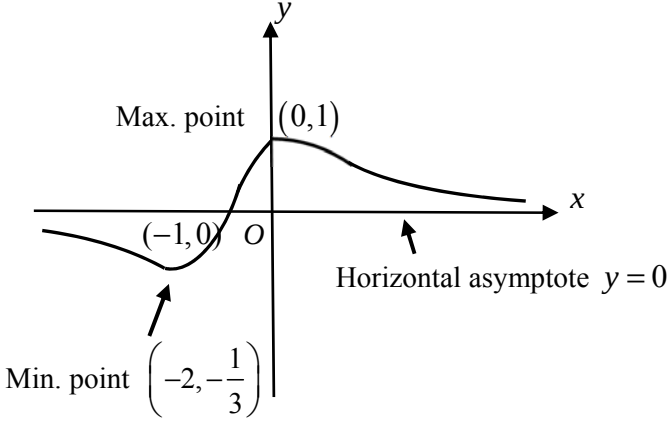


Jurong Junior College
2018 JC2 H2 Mathematics Prelim Paper 1 Solution

Qn	Solution
1	Let \$x and \$y be the price of each daily normal ticket and each daily concession ticket respectively. Booth A : $5nx + ny = 5976$ -----(1) Booth B : $7nx + 2ny = 8712$ -----(2) Booth C : $357x + 51y = 5763$ -----(3) $5x + y - 5976\left(\frac{1}{n}\right) = 0 \quad \text{L L} \quad (1)$ $7x + 2y - 8712\left(\frac{1}{n}\right) = 0 \quad \text{L L} \quad (2)$ $357x + 51y + 0\left(\frac{1}{n}\right) = 5763 \quad \text{L L} \quad (3)$ Or $\frac{5x + y}{7x + 2y} = \frac{5976}{8712} \Rightarrow 1728x - 3240y = 0$ From GC : $x = 15$, $y = 8$, $\frac{1}{n} = \frac{1}{72}$ Each daily normal ticket costs \$15, and each daily concession ticket costs \$8 and $n = 72$.

Qn	Solution	Marks	Remarks
2(i)	Since $x = -1$ is an asymptote, $\Rightarrow d = 1$	[B1]	
	Since $y = x$ is an asymptote, $y = x + \frac{A}{x+1} = \frac{x^2 + x + A}{x+1} = \frac{ax^2 + bx + c}{x+1}$	[M1]	Use long division $y = \frac{ax^2 + bx + c}{x+1}$ $= ax + b - a + \frac{a+c-b}{x+1}$
	$\Rightarrow a = 1$ and $b = 1$	[A2]	A1 for $a = 1$ A1 for $b = 1$
	$\frac{dy}{dx} = 1 - \frac{A}{(x+1)^2}$ when $x = 0$, $\frac{dy}{dx} = 0$ $\Rightarrow A = 1$	[M1]	Or : when $x = -2$, $\frac{dy}{dx} = 0$
	$\Rightarrow c = 1$	[A1]	
(ii)	$y = \frac{x^2 + x + 1}{x+1}$ $y(x+1) = x^2 + x + 1$ $x^2 + (1-y)x + (1-y) = 0$	[M1]	Form a quadratic equation in x .
	For real x , $(1-y)^2 - 4(1)(1-y) \geq 0$	[M1]	Use Discriminant
	$(1-y)^2 - 4(1)(1-y) \geq 0$ $1 - 2y + y^2 - 4 + 4y \geq 0$ $y^2 + 2y - 3 \geq 0$ $(y+3)(y-1) \geq 0$	[M1]	Solve y
	$\Rightarrow \underline{\underline{y \leq -3 \text{ or } y \geq 1 \text{ (ans)}}$	[A1]	
(iii)	 Max. point $(0, 1)$ Min. point $\left(-2, -\frac{1}{3}\right)$ Horizontal asymptote $y = 0$	[G3]	G1 for shape G1 for $y = 0$ & $(-1, 0)$ G1 for $(0, 1)$ & $\left(-2, -\frac{1}{3}\right)$

Qn	Solution	Marks	Remarks
3(i)	$\int \frac{x}{(4+3x^2)^2} dx = \frac{1}{6} \int (6x)(4+3x^2)^{-2} dx$	[M1]	
	$= -\frac{1}{6(4+3x^2)} + c$	[A1]	
(ii)	$u = 2x \quad \frac{dv}{dx} = \frac{x}{(4+3x^2)^2}$ $\frac{du}{dx} = 2 \quad v = -\frac{1}{6(4+3x^2)}$		Integration by parts with the correct u and $\frac{dv}{dx}$.
	$\int_0^{\frac{2}{\sqrt{3}}} \frac{2x^2}{(4+3x^2)^2} dx = \left[-\frac{2x}{6(4+3x^2)} \right]_0^{\frac{2}{\sqrt{3}}} + \frac{1}{3} \int_0^{\frac{2}{\sqrt{3}}} \frac{1}{4+3x^2} dx$	[M1]	
	$= -\frac{1}{12\sqrt{3}} + \frac{1}{3\sqrt{3}} \int_0^{\frac{2}{\sqrt{3}}} \frac{\sqrt{3}}{2^2 + (\sqrt{3}x)^2} dx$ $= -\frac{1}{12\sqrt{3}} + \left[\frac{1}{6\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}x}{2} \right) \right]_0^{\frac{2}{\sqrt{3}}}$	[M1]	Award mark for $\tan^{-1} \left(\frac{\sqrt{3}x}{2} \right)$
	$= \frac{\pi}{24\sqrt{3}} - \frac{1}{12\sqrt{3}}$	[M1]	Substitution of limits to the two anti-derivatives
	$= \frac{\pi - 2}{24\sqrt{3}}$	[A1]	

Qn	Solution	Marks	Remarks
4(i)	$\frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta}$	[M1]	
	$\frac{dy}{dx} = 0 \Rightarrow \sin \theta = 0 \Rightarrow \theta = \pi$	[M1]	Award mark for $\frac{dy}{dx} = 0$ and attempt to solve for θ
	$y = 1 - \cos \pi = 2$	[A1]	
(ii)	At $\theta = \frac{2\pi}{3}$, $x = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$, $y = \frac{3}{2}$, $\frac{dy}{dx} = \frac{\frac{\sqrt{3}}{2}}{1 + \frac{1}{2}} = \frac{\sqrt{3}}{3}$.	[M1]	
	Gradient of normal $= -\sqrt{3}$	[A1]	
	Eqn of normal is $y - \frac{3}{2} = -\sqrt{3} \left[x - \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \right]$ $y = -\sqrt{3}x + \frac{2\sqrt{3}\pi}{3}$ (Shown)	[M1]	Show working. AG
	$x = 0$, $y = \frac{2\sqrt{3}\pi}{3} \Rightarrow Q \left(0, \frac{2\sqrt{3}\pi}{3} \right)$ $y = 0$, $x = \frac{2\pi}{3} \Rightarrow P \left(\frac{2\pi}{3}, 0 \right)$	[M1]	
	Area of triangle $= \frac{1}{2} \times \frac{2\pi}{3} \times \frac{2\sqrt{3}\pi}{3} = \frac{2\sqrt{3}\pi^2}{9}$ units ²	[A1]	
(iii)	$\frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta}$ $\frac{d\left(\frac{dy}{dx}\right)}{d\theta} = \frac{\cos \theta (1 - \cos \theta) - \sin^2 \theta}{(1 - \cos \theta)^2} = \frac{1}{\cos \theta - 1}$	[M1]	Differentiate wrt θ Accept $\frac{d^2y}{d\theta^2} = \frac{1}{\cos \theta - 1}$
	$\frac{d\left(\frac{dy}{dx}\right)}{dt} = \frac{d\left(\frac{dy}{dx}\right)}{d\theta} \cdot \frac{d\theta}{dt} = \frac{1}{\cos \frac{\pi}{3} - 1} (2)$	[M1]	Use chain rule $\frac{d\left(\frac{dy}{dx}\right)}{d\theta} \cdot \frac{d\theta}{dt}$
	$= -4$ units/s	[A1]	

Qn	Solution	Marks	Remarks
5	$C = \pi(3r)^2 k + 2\pi(3r)^2 (2k) + 2\pi rh(2k)$ $= 45\pi r^2 k + 4\pi rhk$	[M1]	
	$\Rightarrow h = \frac{C - 45\pi r^2 k}{4\pi rk}$	[M1]	
	$V = \frac{2}{3}\pi(3r)^3 - \pi r^2 h$ $= 18\pi r^3 - \pi r^2 \left(\frac{C}{4\pi rk} - \frac{45}{4}r \right)$ $= \frac{117}{4}\pi r^3 - \frac{Cr}{4k}$	[M1]	Show working. AG
(i)	$\frac{dV}{dr} = \frac{117}{4}\pi(3r^2) - \frac{C}{4k}$	[M1]	
	At stationary value of V, $\frac{dV}{dr} = 0$ $\frac{117}{4}\pi(3r^2) - \frac{C}{4k} = 0$ $\Rightarrow r^2 = \frac{C}{351\pi k} \Rightarrow r = \sqrt{\frac{C}{351\pi k}}$	[A1]	
(ii)	$\frac{h}{r} = \frac{C - 45\pi r^2 k}{4\pi r^2 k} = \frac{C}{4\pi r^2 k} - \frac{45}{4}$ $= \frac{C}{4\pi k \left(\frac{C}{351\pi k} \right)} - \frac{45}{4}$	[M1]	
	$= \frac{153}{2}$	[A1]	
(iii)	Since $h = 76.5r$ does not satisfy $h < 3r$, $\therefore$ it is not possible for this toy to have a stationary value of V.	[B1]	

Qn	Solution	Marks	Remarks
6(i)	Area of triangle $ABC = \frac{1}{2} \vec{AB} \times \vec{AC} $	[M1]	
	$= \frac{1}{2} (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) $ $= \frac{1}{2} \mathbf{b} \times \mathbf{c} - \mathbf{b} \times \mathbf{a} - \mathbf{a} \times \mathbf{c} + \mathbf{a} \times \mathbf{a} $	[M1]	
	$= \frac{1}{2} \mathbf{b} \times \mathbf{c} + \mathbf{a} \times \mathbf{b} + \mathbf{c} \times \mathbf{a} $	[A1]	
	$\Rightarrow \frac{1}{2} \mathbf{b} \times \mathbf{c} + \mathbf{a} \times \mathbf{b} + \mathbf{c} \times \mathbf{a} = \frac{1}{2} \mathbf{c} - \mathbf{a} \times \text{height}$ shortest distance from B to $AC = \frac{ \mathbf{b} \times \mathbf{c} + \mathbf{a} \times \mathbf{b} + \mathbf{c} \times \mathbf{a} }{ \mathbf{c} - \mathbf{a} }$ (Shown)	[M1]	Show working AG
6(ii)	$\vec{RA} = \frac{1}{3} \vec{BA} = \frac{1}{3} (\mathbf{a} - \mathbf{b})$	[B1]	
	$\vec{RS} = \vec{OS} - \vec{OR} \qquad \vec{RS} = \vec{RA} + \vec{AS}$ $= \left(\frac{2\mathbf{c} + \mathbf{a}}{3} \right) - \left(\frac{2\mathbf{a} + \mathbf{b}}{3} \right) \quad \text{or} \quad = \frac{1}{3} (\mathbf{a} - \mathbf{b}) + \frac{2}{3} (\mathbf{c} - \mathbf{a})$ $= -\frac{1}{3} \mathbf{a} - \frac{1}{3} \mathbf{b} + \frac{2}{3} \mathbf{c} \qquad \qquad \qquad = -\frac{1}{3} \mathbf{a} - \frac{1}{3} \mathbf{b} + \frac{2}{3} \mathbf{c}$	[M1] [A1]	
	$\vec{RA} \cdot \vec{RS} = \frac{1}{3} (\mathbf{a} - \mathbf{b}) \cdot \left(-\frac{1}{3} \mathbf{a} - \frac{1}{3} \mathbf{b} + \frac{2}{3} \mathbf{c} \right)$ $= \frac{1}{9} [(\mathbf{a} - \mathbf{b}) \cdot (-\mathbf{a} - \mathbf{b} + 2\mathbf{c})]$ $= \frac{1}{9} [-\mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{b} + 2(\mathbf{a} - \mathbf{b}) \cdot \mathbf{c}]$	[M1]	
	$= \frac{1}{9} [- \mathbf{a} ^2 + \mathbf{b} ^2 + 2(\mathbf{a} - \mathbf{b}) \cdot \mathbf{c}]$	[M1]	
	$OACB$ is a kite with $OA = OB$, $CA = CB$ and $\vec{BA} \perp \vec{OC}$ $\Rightarrow \mathbf{a} = \mathbf{b}$ and $(\mathbf{a} - \mathbf{b}) \cdot \mathbf{c} = 0$ $\therefore \vec{RA} \cdot \vec{RS} = 0$ (Shown)	[M1]	

Qn	Solution	Marks	Remarks
7(a) (i)	$y = \frac{3x+k}{x-b}$ $xy - by = 3x + k$ $x(y-3) = by + k$ $x = \frac{by+k}{y-3}$ $f^{-1}(x) = \frac{bx+k}{x-3}$	[M1]	
	For $f(x) = f^{-1}(x)$, $\frac{3x+k}{x-b} = \frac{bx+k}{x-3}$ $\therefore b = 3$	[A1]	
	Also, $3x + k \neq m(x-3)$ since f is a one to one function. $\therefore k \neq -9$	[A1]	
(ii)	$fg(x) = 2x - 1$ $g(x) = f^{-1}(2x-1)$ $= f(2x-1)$ $= \frac{3(2x-1)+k}{(2x-1)-3}$	[M1]	
	$= \frac{6x-3+k}{2x-4}, x \neq 2$	[A1]	

7(b) (i)		[G3]	M1: Correct shape, maximum and minimum points on the interval $1 \leq x \leq 4$. A1: y-intercept at (0, 3) and interpreting $h(x+3) = h(x)$ A1: Endpoints at (-4, 0) and (6, 3)
(ii)	$\int_{-1}^2 h(x) dx = \int_2^4 (-x^2 + 8x - 12) dx + \frac{1}{2}(1)(4)$ $= \frac{22}{3}$	[M1]	Finding $\int_{-1}^2 h(x) dx$ or equivalent
	$\int_{-\frac{3}{2}}^3 h(x) dx = \frac{1}{2} \left(\frac{1}{2} \right) (2) + \left(\frac{22}{3} \right) + \int_2^3 (-x^2 + 8x - 12) dx$	[M1]	
	$= \frac{19}{2} = 9.5$	[A1]	

Qn	Solution	Marks	Remarks
8(i)	$\frac{dx}{dt} = kx(1-x)$	[B1]	
	When $x = \frac{1}{3}$, $\frac{dx}{dt} = \frac{1}{6} = k\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)$	[M1]	
	$\therefore k = \frac{3}{4}$ i.e. $\frac{dx}{dt} = \frac{3}{4}x(1-x)$	[A1]	
(ii)	$\int \frac{1}{x(1-x)} dx = \int \frac{3}{4} dt$	[M1]	
	$\int \left(\frac{1}{x} + \frac{1}{1-x} \right) dx = \int \frac{3}{4} dt$	[M1]	Use of partial fractions (or equivalent)
	$\ln x - \ln 1-x = \frac{3}{4}t + c$	[M1]	
	$\ln \left \frac{x}{1-x} \right = \frac{3}{4}t + c$ $\frac{x}{1-x} = e^{\frac{3}{4}t+c}$ $\frac{x}{1-x} = Ae^{\frac{3}{4}t}$, where $A = e^c$	[M1]	
	when $t = 0$, $x = \frac{1}{3}$, $\frac{1}{2} = Ae^0$	[M1]	
	$A = \frac{1}{2}$ i.e. $\frac{x}{1-x} = \frac{1}{2}e^{\frac{3}{4}t}$	[A1]	
	when $t = 2$, $\frac{x}{1-x} = \frac{1}{2}e^{\frac{3}{2}}$	[M1]	
	$2x = e^{\frac{3}{2}} - xe^{\frac{3}{2}}$ $x\left(2 + e^{\frac{3}{2}}\right) = e^{\frac{3}{2}}$ $x = \frac{e^{\frac{3}{2}}}{2 + e^{\frac{3}{2}}}$	[M1]	
	% of crop destroyed = $\frac{e^{\frac{3}{2}}}{2 + e^{\frac{3}{2}}} \times 100 = 69.1\%$	[A1]	

Qn	Solution	Marks	Remarks
9(a)	$z^2 + (i-4)z + (6-2i) = 0$ $z = \frac{-(i-4) \pm \sqrt{(i-4)^2 - 4(6-2i)}}{2}$	[M1]	
	$z = \frac{-i+4 \pm \sqrt{i^2 - 8i+16 - 24 + 8i}}{2}$ $z = 2+i, 2-2i$	[A1]	
9(b)	$w = re^{i\theta}$ and $w^* = re^{-i\theta}$ $p = \frac{re^{i\theta}}{re^{-i\theta}} = e^{i(2\theta)}$	[M1]	
	$ p =1$ and $\arg(p) = 2\theta$	[A1]	
	$p^6 = e^{i(12\theta)} = \cos(12\theta) + i\sin(12\theta)$	[M1]	
	$0 < \theta < \frac{\pi}{2} \Rightarrow 0 < 12\theta < 6\pi$ For p^6 to be real, $\sin(12\theta) = 0$, i.e. $12\theta = \pi, 2\pi, 3\pi, 4\pi, 5\pi$	[M1]	
	For p^6 to be positive $\Rightarrow \cos(12\theta) > 0$ $\Rightarrow 12\theta = 2\pi, 4\pi$ $\theta = \frac{\pi}{6}, \frac{\pi}{3}$	[A1]	
9(c)	2 complex roots.		
(i)	Since $P(z)$ has real coefficients and $z = 1+i$ is a complex root, its conjugate is another root. There cannot be a third complex root since $z = 2$ is a real root.	[B1]	
(ii)	$P(z) = (z-(1+i))(z-(1-i))(z-2)(az+c)$	[M1]	Accept $(z+c)$ Or $P(z) = k(z-(1+i))(z-(1-i))(z-2)(z+c)$
	$P(i) = (i-(1+i))(i-(1-i))(i-2)(ai+c) = 10+10i$	[M1]	
	$(-1)(-1+2i)(i-2)(ai+c) = 10+10i$ $(5i)(ai+c) = 10+10i$ $-5a+5ci = 10+10i$	[M1]	
	Comparing real and imaginary parts, $a = -2$ and $c = 2$	[A1]	
	$\therefore -2z+2=0 \Rightarrow z=1$ Hence the other 2 roots are $z=1-i$ and $z=1$.	[A1]	

Qn	Solution	Marks	Remarks
10 (i)	$\begin{pmatrix} 2 \\ 3 \\ -12 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ -7 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$	[M1]	
	$\begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = 12$	[M1]	
	Cartesian equation of plane $\pi : 3x + 2y + z = 12$	[A1]	
	At support beam $S_1 : y = z = 0 \Rightarrow$ vertex $A = (4, 0, 0)$	[M1]	AG
10 (ii)	$\text{Acute angle of inclination of wall} = \cos^{-1} \frac{\left \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \right }{\sqrt{9+4+1}}$	[M2]	M1 for using formula to find angle between 2 vectors M1 for $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
	$= 74.498^\circ ; 74.5^\circ$ Since $74.5^\circ < 80^\circ$, the safety standard is met.	[A1]	
10 (iii)	$\vec{ON} = \lambda \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ for some λ	[B1]	
	$\Rightarrow \begin{pmatrix} 3\lambda \\ 2\lambda \\ \lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = 12$ $\lambda = \frac{6}{7}$	[M1]	
	Point $N : \left(\frac{18}{7}, \frac{12}{7}, \frac{6}{7} \right)$	[A1]	
	Length of 4th support beam $= \frac{6}{7} \sqrt{9+4+1}$	[M1]	
	$= \frac{6\sqrt{14}}{7}$	[A1]	